electron structure of the atom is said to be built up of a number of individual *shells.* Those chemical elements in which the extranuclear atomic structure is, itself, completed by the completion of such a shell, hold a special significance. They are the *inert gases.* The first one of these, helium, may be imagined, under the particle aspect of matter, as a nucleus with two electrons revolving about it, at approximately equal distances (Figure 8 (*a*)). Thus the first shell is completed by two electrons already. The next element, lithium, has one electron more, and this third electron moves in a much wider orbit, as a lone electron in a new shell (Figure 8 (*b*)). It is evident that this atom can yield up an electron very easily, and consequently, occurs very frequently as a positive ion. This is the explanation of the electro-positive character of lithium, the most important characteristic of the chemical behaviour of this element.

And so on further. After a certain number of elements, we always find a completed shell, as in helium. The chemical significance of this fact is that these elements—the inert gases, helium, neon, argon, krypton, xenon—do not react chemically at all. They always represent the end of a period, as shown in Table III at the end of this book.

As we have stated before, the first period consists of only two elements: hydrogen and helium. The second period contains eight elements: lithium, beryllium, boron, carbon, nitrogen, oxygen, fluorine and neon. Since the element fluorine contains seven electrons in its outermost shell, one more is needed to complete this shell. This fact gives a clue to the chemical properties of fluorine: The fluorine atom always tends to complete this outermost shell by taking up an eighth electron, and thus to have an electro-negative character, and to occur in solutions negatively charged, as a rule. Obviously, those elements which occur at the beginning of a period (hydrogen and the alkali metals), and which therefore will readily give up an electron, combine particularly readily with one of the penultimate elements of the periods, the halogens, which will readily take up an electron. Examples of these combinations are hydrogen fluoride (HF) and sodium chloride, ordinary table salt (NaCl).

